

SKILLS AND COMPETENCIES

LLMs, DL, & ML: PyTorch, Huggingface, Transformers, Accelerate, AWS SageMaker, fastai, Ax, Optuna, scikit-learn, pandas, numpy, spaCy.

ML Engineering: Experimentation, fine-tuning, evaluation pipelines, predictive modeling, scalable deployment.

MLOps, DevOps, CI/CD: MLFlow, Neptune, LightningAI, AWS, GCP, CI/CD pipelines, Docker, Kubernetes, Prometheus, Grafana.

Full-Stack Engineering: FastAPI, OpenAPI, REST APIs, Microservices for inference, Streamlit, D3.js, Test Driven Development (TDD), Front-end frameworks, Back-end frameworks.

Collaboration & Leadership: Mentorship, Code reviews, Agile/Scrum, Research-to-production translation, Strong verbal and written communication skills, Strong work ethic, Self-starter, Problem solver.

EXPERIENCE

R&D Scientist II - Quantum NLP (QNLP)

Quantinuum Ltd.

Oct 2023 – Present

Oxford, United Kingdom

• LLM Fine-tuning for Specialized Reasoning Tasks

- Designed and deployed LLM fine-tuning pipelines (SFT+GRPO), utilizing PEFT-techniques and multi-GPU training within the LightningAI platform, to train Qwen2 models for the task of creating and manipulating string diagrams that required formally verifiable steps or Chain-of-Thought (COT) reasoning.
- Worked with senior researchers to translate their domain knowledge in string diagrams and category theory into the synthetic data generation scripts, serialization/deserialization processes, and prompts to create the training and evaluation datasets.
- Designed custom evaluation metrics to measure the performance of the models on the generation and manipulation tasks.

• Novel BERT model for End-to-End Generation of String Diagrams from Sentences

- Designed, implemented, and trained a custom BERT-based model for converting sentences directly to string diagrams which simplified QNLP training pipelines , complete with LoRA fine-tuning and automated hyperparameter tuning using Ax for more efficient training, and experiment and artifact tracking using both Neptune and MLFlow.
- Implemented a compact tree-based representation of string diagrams to create the training and evaluation datasets and an appropriate data augmentation strategy for the task.
- Mentored interns who worked on this project and worked with them closely so they can contribute to the new features being developed, while maintaining good codebase quality for ease of reproducibility, maintainability, and extensibility.

• Translation of Theoretical Research to Practical Implementations

- Worked with senior researchers on an improved implementation of the DisCoCirc paper  which describes a compositional, non-LLM-based model for how texts derive meaning from their constituent sentences.
- Created benchmarking datasets to compare the performance of the new implementation versus the original implementation .
- Wrote reusable, configurable scripts to check the coverage of the new implementation on the dataset, which simplified the testing pipeline and increased the confidence with the new pipeline and detailed these results in an accompanying preprint.
- Implemented visualization functions to allow inspection of the output circuit diagrams.
- Presented work to other colleagues not necessarily familiar with QNLP.

• lambeq Research and Development

- Implemented new features and bug fixes for better usability when training QNLP models.
- Simplified and standardized parts of the API.
- Determined bottlenecks in the pipeline by benchmarking each of its steps, proposed and implemented improvements.
- Performed regular code reviews with the team to enforce best practices.

- Replicated and built upon the results of past research papers for miscellaneous areas of interest.

Data Scientist

AI Cross Inc.

Nov 2021 — Sep 2023

Tokyo, Japan

• Machine Learning Platform-as-a-Service 🔗

- Designed and implemented the core ML module of the platform that allowed users to train models and retrieve inference results via APIs.

• Business Marketing Platform-as-a-Service 🔗

- Developed prototype for generating and delivering AI-based recommendations in 1 month, which enabled the team to pitch and to present to potential clients, and to validate the value provided by the service.
- Created the system architecture for the product based on the initial prototype.
- Implemented the data ingestion, model training, and recommender components of the system, working across the entire stack (frontend, backend, ML module).
- Constantly communicated with the sales team to understand client needs and re-prioritized the development milestones accordingly.
- Implemented A/B testing feature that allowed clients to experiment with their strategies and choose the best option within statistical bounds.
- Self-learned about UI/UX and implemented changes to the design of the platform.
- Operationalized the system, closely collaborating with our CTO in the process.

Machine Learning Engineer

LINE Fukuoka Corporation

Sep 2019 — Nov 2021

Fukuoka, Japan

• Detection of High-Profile Individuals from Caricatures

- Benchmarked the existing tool with reusable and configurable Python scripts.
- Developed an image detection pipeline based on the WarpGAN paper.
- Implemented Spark, Airflow batch-processing pipeline that handled 1M image/s throughput.
- Developed and deployed APIs for real-time inference and visualization of batch detection results.
- Added service monitoring via Prometheus and Grafana to tighten the feedback loop in detecting target events.
- Wrote the entire CI/CD pipeline that allowed for faster iteration.

• Content Suggestion via Image-Based Emotion Recognition

- Coordinated with the product team to scope the requirement.
- Designed the model and solution architecture based on existing literature on emotion recognition and recommendation systems.
- Trained baseline models for recognizing emotion from input images.
- Developed and deployed APIs for real-time inference.
- Implemented service monitoring via Prometheus and Grafana.

• Machine Learning Tooling and Knowledge-Sharing Tasks

- Improved existing and created new ML/MLOps tools with the team.
- Gave presentations on the following areas of interest: meta-learning, model quantization 🔗, auto-ML 🔗, causal inference 🔗, secure AI 🔗, and explainable AI 🔗.

Data Scientist

Teradata Corporation

Nov 2016 — Sep 2019

Manila, Philippines

• Survey Response Classification

- Trained LSTM + word-embedding models using Keras with Tensorflow and scikit-learn for the classification of survey responses to easily diagnose problems the end-users are facing.
- Designed the architecture of the end-to-end (E2E) system with the principal data scientist.
- Developed the app for data ingestion and prediction visualization using R, Python, and SQL that reduced manual labor and errors done by client.
- Wrote a Docker pipeline to streamline both local and production deployment.
- Hosted learning sessions for new technologies for the rest of the team.

Senior Full-Stack Engineer

icannhas Inc.

Sep 2011 — Nov 2016

Manila, Philippines

EDUCATION

Master of Science in Physics

2021 – Present

University of the Philippines Diliman, Quezon City (On leave)

Bachelor of Science in Civil Engineering, *magna cum laude*

2006 – 2011

University of the Philippines Diliman, Quezon City

Awardee, Engineering Top 100 (out of 5000 students) in Academics, 2011

10th Place, ACM International Collegiate Programming Contest Asia Manila Regional Contest, 2009

Honorable Mention, 37th International Physics Olympiad, 2006